



# Exploring the Effects of Preferential Sampling and Delayed Data on Real-Time Infectious Disease Modeling

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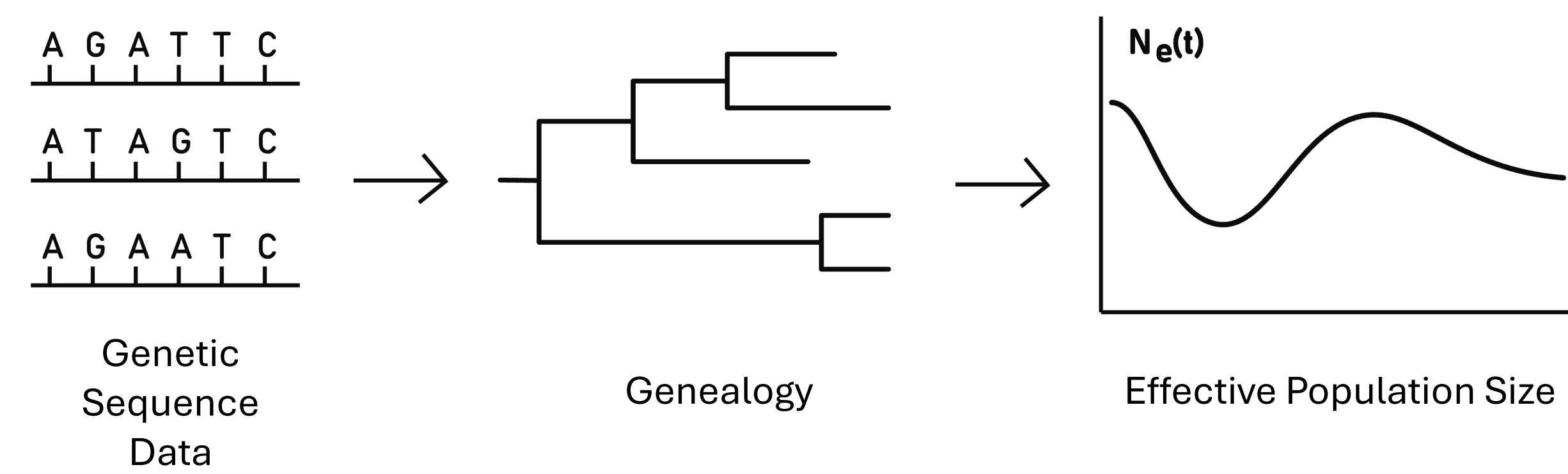


## Introduction

- Knowledge of how many people are infected with a given disease is imperative for informed public health response, but is often difficult to ascertain
- Pathogens rapidly mutate, so the genetic diversity of the pathogen in infected individuals can inform about how the disease spread through a population
- Phyldynamic models utilize genetic data to infer dynamics about a pathogen, but there is no single model that is "best" for all scenarios
- In this work we used simulations to compare the performance of common phyldynamic models for real-time inference

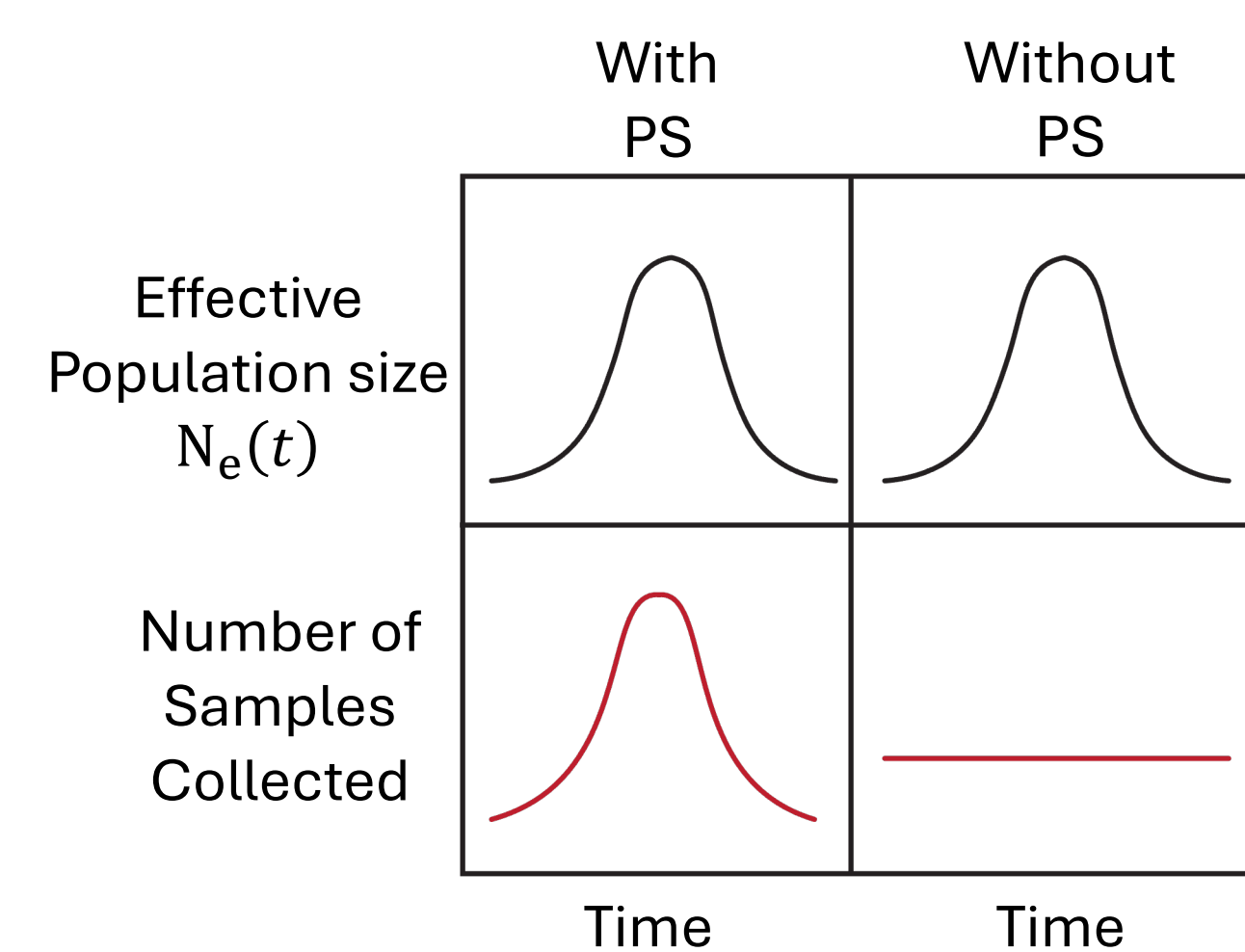
## Inference Challenges

### Phyldynamics



- Methodology utilizes pathogen genetic sequences as data
- These samples are collected over time, and the mutational differences are used to construct a genealogy tree through coalescent modeling
- This evolutionary tree describes the virus's lineage, which is a powerful tool when investigating pathogen dynamics
- Phyldynamics can be used to infer effective population size, which can inform about the number of people infected or transmission, under certain conditions

### Problem 1: Preferential Sampling



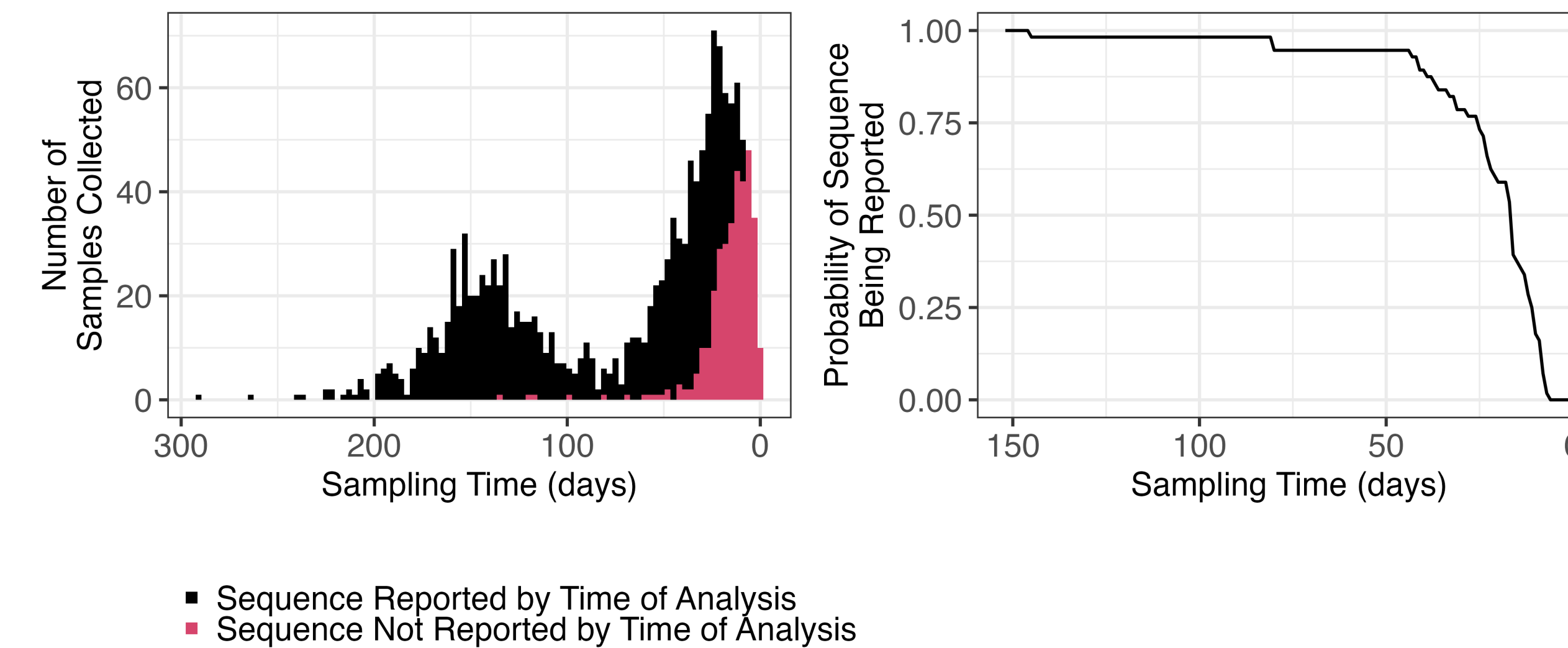
Illustrative example depicting PS relationship

Preferential sampling (PS) is when the rate at which samples are collected is related to the number of people infected

- E.g., sampling more often when more people are infected
- The sampling times can provide information when PS occurs
- Accounting for PS in phyldynamic modeling can lead to more accurate and precise estimates of the effective population size<sup>[3]</sup>

## Inference Challenges Continued

### Problem 2: Reporting Delays



The time between when the blood sample is collected, sequenced, and available for analysis is called the reporting delay

- In real-time analysis, reporting delays cause a missing-data problem
- Intuitively, missing data near present time may lead to underestimation, but this is understudied across common phyldynamic methods
- Currently, only one model corrects for reporting delays

## Methods

### Bayesian Posterior Model

$$\Pr(N_e(t) | \text{Data})$$

### Model Breakdown

$$\propto \Pr(\text{Genealogy} | N_e(t), \text{Data}) \Pr(N_e(t) | \text{Data})$$

These models differ in how the sampling intensity is modeled

### Bayesian Nonparametric Phyldynamic Reconstruction Model<sup>[5]</sup> (BNPR)

$\lambda(t)$  is assumed independent from  $N_e(t)$

### Preferential Sampling BNPR Model<sup>[3]</sup> (BNPR PS)

$$\log(\lambda(t)) = \beta_0 + \beta_1 \log(N_e(t))$$

### Reporting Delay-Aware Preferential Sampling BNPR Model<sup>[4]</sup> (Delay-Aware BNPR PS)

$$\log(\lambda(t)) = \beta_0 + \beta_1 \log(N_e(t)) + \log(r(t))$$

### Adaptive Preferential Sampling BNPR Model<sup>[1]</sup> (BNPR Ada PS)

$$\log(\lambda(t)) = \log(\beta(t)) + \log(N_e(t))$$

With

$\lambda(t)$  = observed sampling intensity

$r(t)$  = probability of a sample being reported at time  $t$

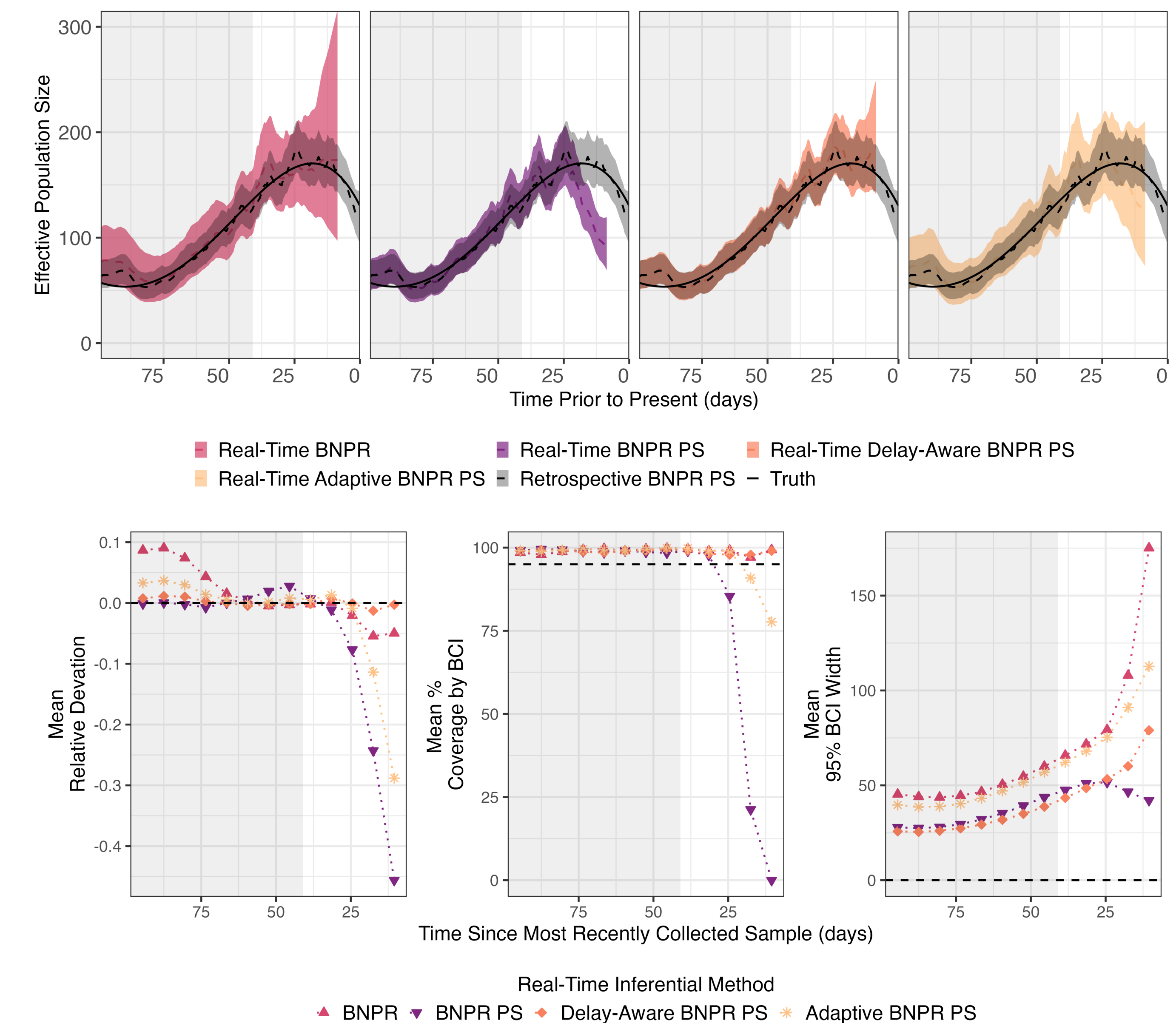
$N_e(t)$  = effective population size "measure of genetic diversity of the pathogen"

$\beta_0, \beta_1, \beta(t)$  = model specific preferential sampling parameters

## Results From Simulation Studies

When reporting delays are present and preferential sampling is constant:

- Delay-Aware BNPR PS model is the most accurate and precise method
- BNPR Ada PS model performs reasonably well, despite not accounting for reporting delays, but accuracy severely decreases near present time
- BNPR PS model lacks accuracy and BNPR model lacks precision



## Discussion

### Novelty of This Work:

- Compared and contrasted these models in different realistic scenarios
  - Specifically, the investigation of how the Adaptive Preferential Sampling BNPR Model performs in the presence of reporting delays
- Created tutorials and a function to aid in the adoption and use of these methods, to be added to the R package: **phylodyn2**<sup>[2]</sup>

### Future Directions:

- Further simulations to investigate additional scenarios of inconsistent preferential sampling
- Add correction for reporting delays in BNPR Ada PS Model

## References

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